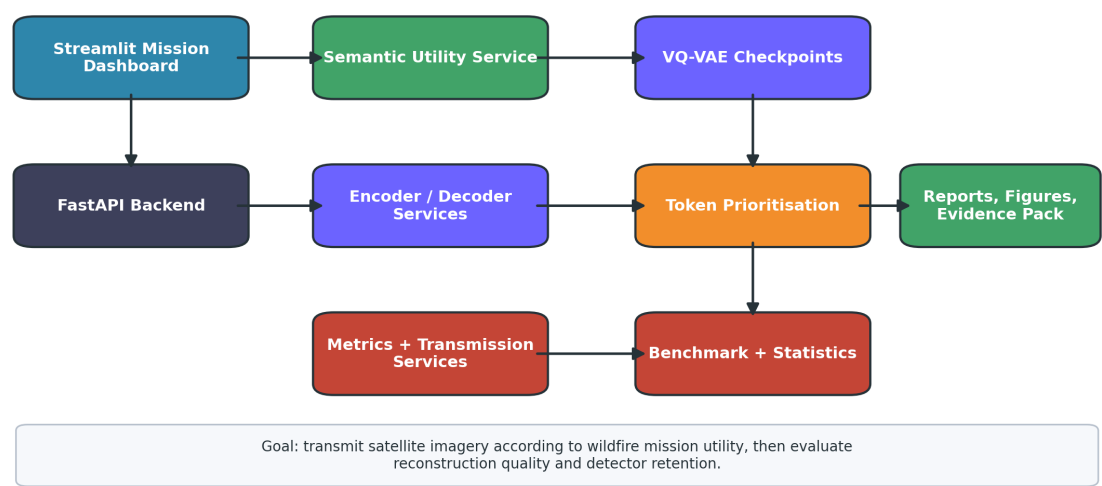


PhD Supervisor Package

Semantic Utility-Aware Compression for Wildfire-Centric Earth Observation Systems

CompressAI System Architecture



Purpose: provide a concise research package for supervisors in Earth Observation, onboard AI, semantic communication, and neural compression.

1. Research Summary

This project investigates whether satellite imagery can be compressed and transmitted according to mission relevance rather than pixel-level distortion alone. The current wildfire-focused system ranks learned visual tokens by semantic utility and evaluates whether wildfire-relevant information survives compression and reconstruction.

Core hypothesis

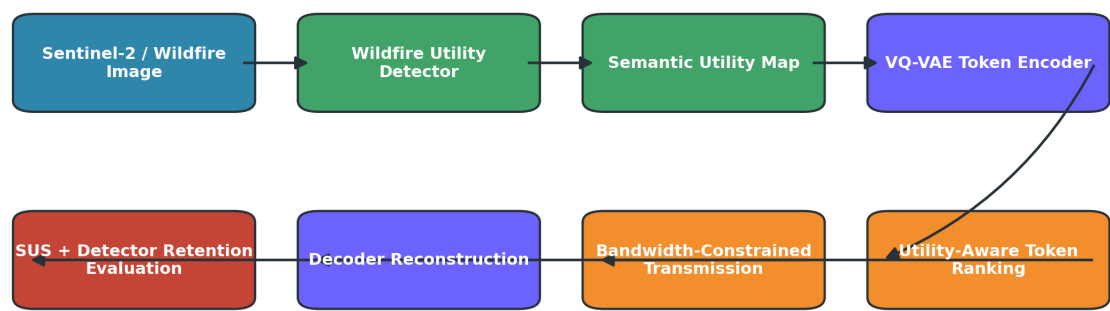
Semantic utility-aware token prioritisation preserves wildfire-relevant Earth Observation information more effectively than conventional distortion-optimised compression under equivalent bandwidth constraints.

Research gap

Existing image compression methods mainly optimise distortion, perceptual quality, or bitrate. Earth Observation workflows often evaluate downstream analytics separately. This project connects utility estimation, learned token compression, satellite communication constraints, and detector-retention evaluation in one experimental platform.

2. Methodology Pipeline

Working Pipeline: Semantic Utility-Aware Compression

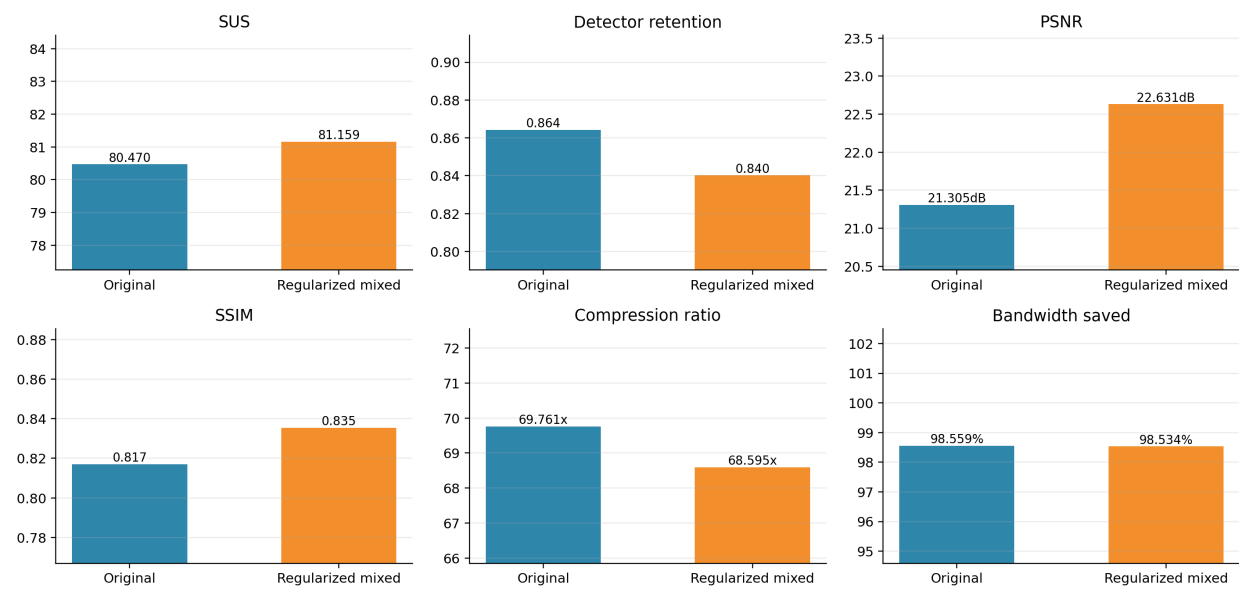


The pipeline converts wildfire and Sentinel-2 imagery into learned VQ-VAE tokens, ranks those tokens according to semantic mission utility, transmits only the selected token subset, reconstructs imagery, and evaluates both reconstruction quality and semantic utility retention.

3. Current Experimental Evidence

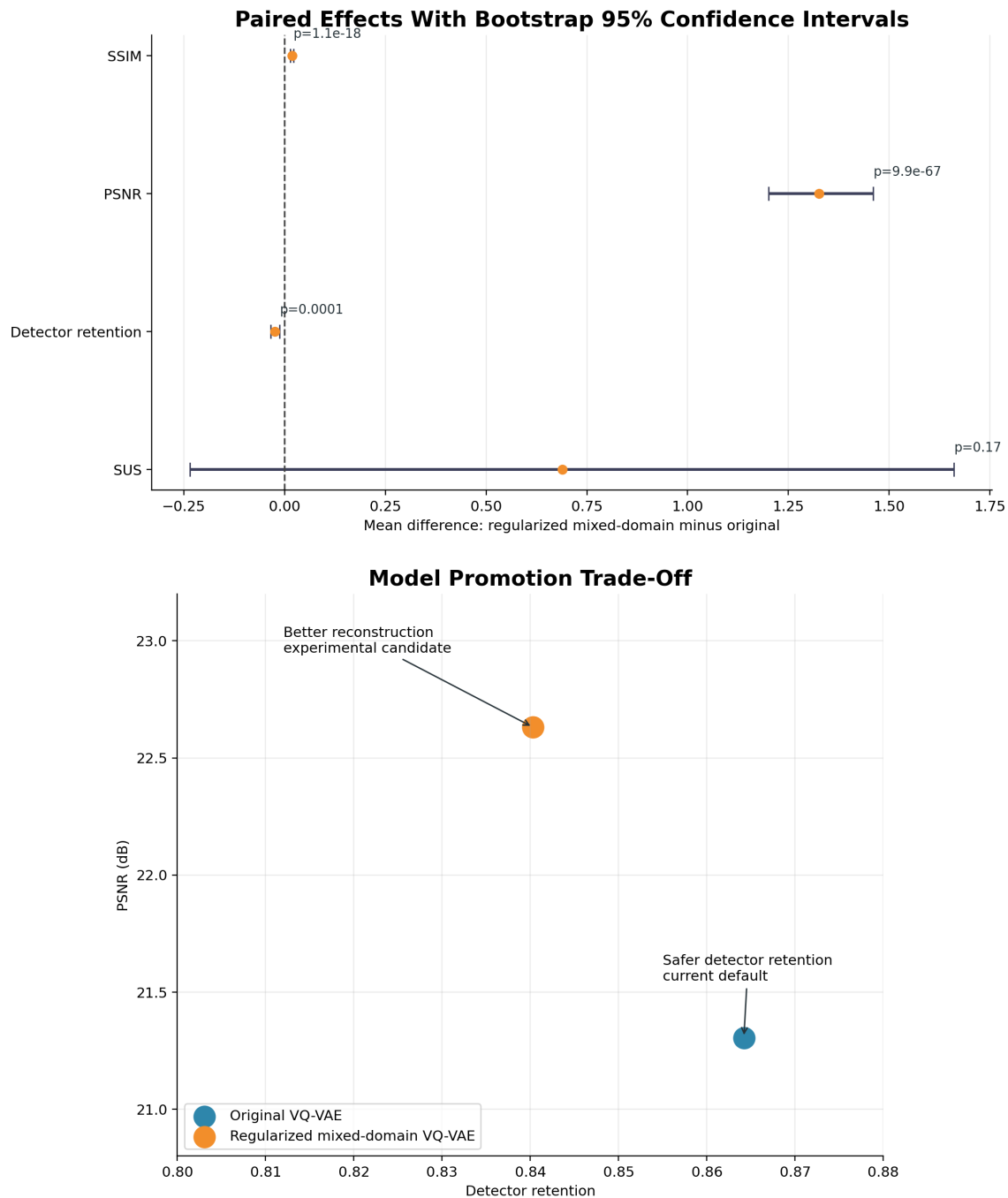
Model	SUS	Detector retention	PSNR	SSIM	Compression ratio	Bandwidth saved
Original VQ-VAE	80.470	0.8642	21.305	0.8170	69.76x	98.56%
Regularized mixed-domain VQ-VAE	81.159	0.8403	22.631	0.8354	68.60x	98.53%

Sentinel-2 500-Patch Benchmark: Original vs Regularized VQ-VAE



Interpretation: satellite-domain fine-tuning improves PSNR and SSIM, but detector retention drops. This is scientifically useful because it shows that visual reconstruction quality and mission utility are not identical objectives.

4. Statistical Result and Model Decision



Decision: the original VQ-VAE remains the conservative default for mission-utility experiments because it preserves detector retention better on the 500-patch Sentinel-2 benchmark. The regularized model is retained as an experimental reconstruction-improved candidate.

5. Proposed PhD Direction

Year	Research focus	Expected output
1	Formalise wildfire utility metrics, improve datasets, validate detector-retention evaluation	Workshop paper and reproducible benchmark
2	Develop utility-preserving token selection and detector-retention-aware representation learning	Conference paper and model comparison
3	Evaluate under realistic satellite communication constraints and cross-dataset EO settings	Journal paper and satellite simulation study

4	Optimise for onboard deployment and finalise thesis framework	Thesis, open-source release, final journal submission
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6. Supervisor Fit

- Earth Observation and remote sensing: wildfire-centric Sentinel-2 validation and mission-oriented evaluation.
- Onboard AI and edge computing: token pruning and compression under bandwidth constraints.
- Semantic communication: transmission of task-relevant information rather than all pixels.
- Neural compression: VQ-VAE tokenisation and reconstruction-quality versus utility trade-offs.

Best next outreach asset: send this package with the GitHub repository, the two-page research summary, and a short demo video.